

Effects of Keishi-ka-jutsu-bu-to (traditional herbal medicine: Gui-zhi-jia-shu-fu-tang) on in vivo insulin action in streptozotocin-induced diabetic rats

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Abstract

This study investigated the effects of the traditional herbal medicine, Keishi-ka-jutsu-bu-to (KJT) on insulin action in vivo and insulin signaling in skeletal muscle in STZ-induced diabetes. Rats were divided into single and 7-days oral administration groups. Euglycemic clamp (insulin infusion rates: 3 and 30 mU/kg/min) was used in awaked rats and the insulin signaling in skeletal muscle was evaluated. At low-dose insulin infusion, the decreased metabolic clearance rates of glucose (MCR) in diabetic rats were improved by a single and 7-days administration of KJT (800 mg/kg BW, p.o.; acute effect: 6.7 ± 0.6 vs. 12.3 ± 1.2 , and 7-days effect: 6.3 ± 0.5 vs. 13.9 ± 1.0 ml/kg/min, $P < 0.001$, respectively). During high-dose insulin infusion, the MCR was increased in 7-days KJT treated diabetes compared with saline diabetes, but, these changes were not observed after a single KJT treatment. About 90% of the increasing effect in MCR induced by the 7-days KJT treatment was blocked by L-NMMA. However, no further additive effects were seen in KJT + SNP treatment. IR β protein increase and decreased IRS-1 protein expression in diabetes were significantly improved by KJT treatment. KJT had no effect on the GLUT4 protein content. The increased tyrosine phosphorylation level of IR β , IRS-1, and IRS-1 associated with PI 3-kinase were significantly inhibited in

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KJT treated diabetes. The present study suggests that the improvement of impaired insulin action in STZ-diabetes by administration of KJT may be due, at least in part, to enhanced insulin signaling, which may be involved with production of nitric oxide (NO).

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Introduction

Diabetic neuropathy is a most common and troublesome complication of diabetic mellitus [50], leading to great morbidity and mortality and resulting in a huge economic burden for diabetes care [19]. Diabetic neuropathy remains an important clinical problem worldwide. Recently, the traditional Chinese medicine (TCM) (called Kampo medicine in Japan) has revalued its efficacy and had an increase in popularity around the world [8,33] because of the supposedly less frequent side effects when compared with modern Western medicine [33]. The TCM, which is a combined prescription, has unique synergistic and synthetic effects that result from interactions between individual herbal components and may induce a wide range of therapeutic potential and utility [8,33]. For example, Keishi-ka-jutsu-bu-to (KJT), one of the traditional herbal medicines, has been prescribed over hundreds of years and is composed of seven crude herbs and could induce vasodilatation, analgesics, and has anti-inflammatory action [49]. KJT has been used for the treatment of various diseases, including neuropathy, arthralgia, and rheumatism [49]. Furthermore, it has been reported that KJT could relieve the subjective symptoms associated with neuropathy like pain, numbness, and cold sensation in the extremities [2]. In addition, previous studies showed that KJT possess vasodilator and platelet anti-aggregator actions [3,44]. In basic experimental studies, KJT inhibited aldose reductase in rat lens and sorbitol accumulation in human erythrocytes [1] and had potent neuromuscular blocking effects in diabetic mice [22]. However, to our knowledge, there are no reports on whether KJT improves insulin action in diabetes.

Nitric oxide (NO) is an important endogenous vasodilator that mediates various physiological functions. Decreased nitric oxide synthase (NOS) activity could contribute to the genesis of many pathological conditions, such as diabetic vasculopathy and neuropathy [10,39,42]. Additionally, according to Chandra et al. 2002 [11] NO maintains aldose reductase in an inactive state and increased NO availability may be a useful strategy for inhibiting the polyol pathway. The above findings raised us a question: may the pharmacological mechanism of KJT be related to the NO? Previous studies indicated that NO was involved in the regulation of skeletal muscle hemodynamics and glucose utilization through vasodilatation associated with insulin [46]. Petrie et al. 1996 [38] reported that the insulin-mediated peripheral tissues glucose uptake in vivo was NO dependent. To evaluate our hypotheses, we first determined whether impairment in insulin action in streptozotocin-induced diabetic rats was improved by KJT treatment using the euglycemic clamp technique. Based on preliminary results, our second objective was to investigate whether the expected improvement in insulin action in diabetic rats given by KJT treatment was mediated by enhanced NO production. Moreover, we further studied the effects of KJT on the molecular mechanisms of insulin signaling: insulin receptor β -subunit (IR β), insulin receptor substrate-1 (IRS-1), phosphatidylinositol 3-kinase

(PI 3-kinase), and glucose transporter 4 (GLUT4) in the skeletal muscle harvested from experimental rats.

Materials and methods

Materials

Male Wistar rats aged 8 weeks (220–230 g) were purchased from CLEA (Japan). Streptozotocin (STZ) and sodium nitroprusside (SNP) were purchased from Sigma (USA). N^G-monomethyl-L-arginine (L-NMMA) was obtained from Calbiochem (Germany), and neutral insulin from Novo Nordisk (Denmark). Anti-PI 3-kinase, anti-IR β , anti-IRS-1, anti-phosphotyrosine, and anti-GLUT4 antibodies were purchased from Santa Cruz Biotechnology (USA). Hyperfilm was purchased from Amersham (UK). All other reagents were of biochemical grade. Herbal extract of KJT was kindly provided by Tsumura Co. (Tokyo, Japan).

Animal care and experimental design

The rats had free access to a standard diet and water and were kept in an environmentally controlled room at 23 ± 1 °C with an alternating 12-h light/dark cycle (light 8:00–20:00). After a 1-week acclimatization period, rats were randomly divided into two groups: non-diabetic control and diabetic groups. Diabetes was induced by a single STZ (50 mg/kg BW, dissolved in sodium citrate buffer: 50 mM, pH 4.5, freshly prepared) injection from the tail vein; control animals were injected with citrate buffer alone. Blood samples were taken from a tail cut for measurement of postprandial blood glucose (PBG, measured 2 hours after feeding) at the beginning of the experiment. Animals with PBG levels greater than 16.6 mM 72 hours after STZ injection were considered diabetic. Thereafter, animals were divided into the following subgroups: acute oral administration group and 7-days repeated oral administration group. Furthermore, the acute administration and the 7-days oral administration rats were further divided into four subgroups and six subgroups, respectively, as shown in Table 1.

KJT and saline administration: KJT was composed of seven medicinal plants (Table 2), extracted with hot water, filtered, lyophilized, and then stored at 4 °C. In the acute oral administration groups (non-diabetic and diabetic rats), a single dose of 800 mg/kg BW, approximately 10-fold higher than that used in humans [20,48], dissolved in 5 ml/kg saline, was administered orally through a gastric tube before the euglycemic clamp and the remaining rats were given with saline only. Repeated oral administration groups were treated once a day for 7-days with KJT (a dose of 800 mg/kg BW) or saline (Table 1).

Surgical procedures: the animals (non-diabetic and STZ-diabetic) were anesthetized with sodium pentobarbital (50 mg/kg BW). The right jugular vein and the left carotid artery were then exposed and isolated through a middle neck incision. Thin silastic catheters were inserted 30 mm into the right jugular vessel, and the cephalic end of the jugular was ligated. Thereafter, the catheters were inserted 25 mm down the left carotid artery vessel, and sutured in situ. After patency was checked, the catheters were flushed with 300 μ l of 0.9% NaCl solution containing heparin (40 U/ml) and then 500 μ l sodium penicillin G (10000 U/ml) was injected. The catheters were then filled with a

Table 1
Time-table for the experimental design

Groups	STZ treatment	Surgical procedures	Medicine administration								Euglycemic clamp	Dissection
Acute oral administration		3 days	#	7 days								
Non-diabetic												
Saline treatment (7)	(-)	(+)	(-)	(-)	(-)	(-)	(-)	(-)	(-)	(-)	⇓ (+)	(+)
KJT treatment (6)	(-)	(+)	(-)	(-)	(-)	(-)	(-)	(-)	(-)	(-)	⇓ (+)	(+)
Diabetic												
Saline treatment (7)	(+)	(+)	(-)	(-)	(-)	(-)	(-)	(-)	(-)	(-)	⇓ (+)	(+)
KJT treatment (7)	(+)	(+)	(-)	(-)	(-)	(-)	(-)	(-)	(-)	(-)	⇓ (+)	(+)
7-days repeated oral administration		3 days	#	7 days								
Non-diabetic												
Saline treatment (6)	(-)	(+)	(+)	(+)	(+)	(+)	(+)	(+)	(+)	(+)	(+)	(+)
Diabetic												
Saline treatment (6)	(+)	(+)	(+)	(+)	(+)	(+)	(+)	(+)	(+)	(+)	(+)	(+)
KJT treatment (6)	(+)	(+)	(+)	(+)	(+)	(+)	(+)	(+)	(+)	(+)	(+)	(+)
KJT treatment + L-NMMA (5)	(+)	(+)	(+)	(+)	(+)	(+)	(+)	(+)	(+)	(+)	⇓ (+)	(+)
Saline treatment + SNP (5)	(+)	(+)	(+)	(+)	(+)	(+)	(+)	(+)	(+)	(+)	⇓ (+)	(+)
KJT treatment + SNP (5)	(+)	(+)	(+)	(+)	(+)	(+)	(+)	(+)	(+)	(+)	⇓ (+)	(+)

The number of rats is shown in parentheses.

: Surgical procedures were performed three days after STZ treatment.

⇓ : Only one oral administration of saline or KJT just before the clamp.

⇓ : L-NMMA and SNP were administered intravenously only once during the clamp procedure.

viscous solution of 50% polyvinylpyrrolidone (PVP) and capped. The free ends of the catheters were attached to short segments of steel tubing and tunnelled subcutaneously around the side of the neck to the top of the skull where they were threaded through a skin incision to the exterior. Animals were then allowed to resume their normal behaviour. Animal care was conformed to the standards set by the “Guidelines for Animal Experimentation” (Nagoya University, 1988).

Euglycemic clamp procedure: the animals were allowed to recover from surgery for 7-days. The clamp studies were then undertaken in the morning after an overnight fast (12-h). Throughout the study, each rat was allowed to move freely within the cage. The PVP solution was aspirated from

Table 2
Composition of KJT

Component	%
Cinnamomi Cortex	20.51
Paeoniae Radix	20.51
Glycyrrhizae Radix	10.25
Zizyphi Fructus	20.51
Zingiberis Rhizoma	5.12
Atractylodis Lanceae	20.51
Aconiti Tuber	2.56

The KJT used in this study was a unprepared bulk powder.

the catheters, and the venous catheter was then used for the infusion of glucose (20% solution) and insulin by infusion pumps (Terumo, Japan). The blood samples for glucose (25 μ l per sample) and hormone (0.3 ml) determinations were drawn from the arterial cannula. The total blood loss during the experimental procedure did not exceed 1.0 ml (the cell fraction was resuspended in saline and returned to the rats in order to prevent reduction in the hematocrit levels). In the clamp studies, the insulin infusions were given at rates of 3.0 (low-dose, 0–90 min) and 30.0 (high-dose, 90–180 min) mU/kg/min. The blood glucose concentrations of diabetic rats were lowered to 7.8 mM by the low-dose insulin infusion and then clamped at this concentration by periodic adjustment of the glucose infusion rate. During the euglycemic clamp, blood glucose concentrations were measured every 10 min and the euglycemia of diabetic and normal rats was maintained at around 7.7 ± 0.3 mM and 3.9 ± 0.1 mM, respectively. The total amount of the infused glucose was considered to be taken up by the body tissues and, under these steady-state conditions of euglycemia and hyperinsulinemia, glucose input was equal to glucose utilization. The glucose infusion rate (GIR) was calculated every 10 min during the clamp study. Metabolic clearance rate for glucose (MCR, ml/kg/min) was then obtained from GIR divided by the corresponding blood glucose concentration. The MCRs from 60 to 90 min and from 150 to 180 min for the euglycemic clamp were regarded as indices of insulin action in peripheral tissues, since a *plateau* glucose infusion rate was achieved during these periods, as reported previously [20,26]. During the low-dose insulin infusion, L-NMMA was infused with a microinjection pump (Terumo, Japan) in KJT treated diabetic + L-NMMA rats; SNP was given to the saline treated diabetic and KJT treated diabetic + SNP rats, respectively, according to the same protocol described above (Table 1). The dosages for an intravenous administration of L-NMMA (NOS inhibitor) and SNP (NO donor) were 1 mg/kg/min and 15 ng/kg/min, respectively. These dosages were expected to be sufficient to effectively inhibit NOS [16] or elicit insulin action [25]. After the euglycemic clamp study was over, the rats were anesthetized with sodium pentobarbital (50 mg/kg BW), and the gastrocnemius muscles were excised, immediately freeze-clamped at liquid nitrogen temperature, and stored at -80 °C until analysis.

Blood assays

Blood glucose concentration was determined using a glucose analyzer (model 2300; Yellow Springs Instrument, OH). Plasma insulin was assayed with a radioimmunoassay kit (Phadesepa Insulin RIA, Pharmacia AB, Sweden).

Western blotting analysis of IR β , IRS-1, PI 3-kinase, and GLUT4 proteins

The protein expression of the insulin-signaling in the skeletal muscle was measured by the Western blotting method, as previously described [30,31]. Briefly, frozen samples were homogenized using a Polytron homogenizer. The homogenate was kept on ice temperature for 1 hour and then centrifuged at 38,000 rpm at 4 °C for 1 hour. The supernatant proteins (40 μ g) of each sample were size-fractionated by SDS-PAGE (6% or 10% acrylamide gel). The membranes were then incubated with the anti-IR β , anti-IRS-1, anti-PI 3-kinase or anti-GLUT4 antibodies overnight at 4 °C. Bound antibodies were detected by incubation with Goat Anti-Rabbit IgG for 1 hour at room temperature. After washing, blotted proteins were visualized using the Western blotting detection system (ECL Plus, Amersham) and then exposed to

Hyperfilm. Quantification of the band intensity was performed using the public domain NIH image program.

Immunoprecipitation and Western blotting analysis of IR β and IRS-1 tyrosine phosphorylation and IRS-1 associated with PI 3-kinase

The supernatants containing equal amounts of protein (1 mg/ml for each tube) were incubated overnight at 4 °C with anti-IR β (5 mg/ml), or anti-IRS-1 (5 mg/ml), and then with 20 μ l of protein A agarose beads at 4 °C for 4 hours. The immune complexes were washed as described by Goodyear et al., 1995 [18]. Samples were re-suspended in treatment buffer containing β -mercaptoethanol and boiled for 5 min. Phosphorylated proteins were separated by SDS-PAGE (6% or 10% acrylamide gel). Membranes were incubated with phosphotyrosine antibody or PI 3-kinase antibody. Phosphorylated proteins were visualized by the same method described above.

Statistical analysis

All values are expressed as means $\pm$ S.E. Data were analyzed by the one-way analysis of variance. When a significant difference was found ($P < 0.05$), the results were further compared with the Fisher's PLSD test. The StatView 5.0 software (SAS Institute Inc., Cary, NC) was used for the statistical analysis.

Table 3

Body weight and concentrations of blood glucose and plasma insulin before and immediately after the euglycemic clamp procedure at the low-dose and the high-dose insulin infusion

Group	Body wt (g)	Glucose (mM)			Insulin (pM)		
		Basal	Low-dose (90 min)	High-dose (180 min)	Basal	Low-dose (90 min)	High-dose (180 min)
<i>Single administration</i>							
Non-diabetic							
Saline treated (7)	258 ± 4	3.9 ± 0.1	4.1 ± 0.2	4.0 ± 0.1	57 ± 4	178 ± 15	2740 ± 308
KJT treated (6)	256 ± 6	3.8 ± 0.2	3.9 ± 0.1	3.8 ± 0.1	54 ± 6	187 ± 13	2654 ± 188
Diabetic							
Saline treated (7)	235 ± 4	11.1 ± 0.7 ^a	7.8 ± 0.2	7.9 ± 0.1	24 ± 3 ^b	169 ± 8	2573 ± 196
KJT treated (7)	237 ± 5	11.4 ± 0.9 ^a	7.7 ± 0.3	7.9 ± 0.3	26 ± 5 ^b	173 ± 11	2670 ± 276
<i>7-days administration</i>							
Non-diabetic							
Saline treated (6)	253 ± 5	3.8 ± 0.1	3.9 ± 0.1	3.9 ± 0.2	59 ± 7	186 ± 11	2423 ± 215
Diabetic							
Saline treated (6)	226 ± 4	11.3 ± 0.1 ^a	7.8 ± 0.2	7.9 ± 0.2	30 ± 7 ^b	174 ± 21	2272 ± 230
KJT treated (6)	230 ± 6	10.7 ± 0.3 ^a	8.1 ± 0.1	8.0 ± 0.1	32 ± 6 ^b	190 ± 25	2478 ± 301
KJT treated + L-NMMA (5)	228 ± 5	10.8 ± 0.2 ^a	7.9 ± 0.1	7.9 ± 0.2	34 ± 8 ^b	176 ± 19	2470 ± 318
KJT treated + SNP (5)	223 ± 3	11.2 ± 0.1 ^a	7.7 ± 0.2	7.7 ± 0.2	29 ± 4 ^b	198 ± 23	2675 ± 298
KJT treated + SNP (5)	232 ± 7	10.9 ± 0.3 ^a	7.8 ± 0.3	8.1 ± 0.3	31 ± 1 ^b	201 ± 28	2785 ± 327

Values are means $\pm$ S.E. The number of rats is shown in parentheses.

^a Indicates a $P < 0.001$ vs. normal control.

^b Indicates a $P < 0.05$ vs. normal control.

Results

Body weight, blood glucose, and plasma insulin levels

The body weight, blood glucose, and plasma insulin levels before and immediately after the euglycemic clamp are shown in Table 3. Body weight levels and plasma insulin concentrations were significantly lower in diabetic rats compared with the controls. 7-days treatment with KJT did not affect the body weight and insulin levels throughout the experimental period. The blood glucose levels on the 7-days KJT treated diabetes tended to be lower than those of saline treated diabetes, but not significantly. During the low-dose and the high-dose insulin clamp procedures, steady state plasma insulin concentrations were not significantly different among all experimental groups.

Acute effect of KJT on MCRs

In the low-dose insulin clamp studies, the diabetes treated with a single KJT administration showed significantly higher MCRs than those of the saline-treated diabetes ($P < 0.001$). Thus, the decreased MCR in diabetic rats was partially improved by a single KJT treatment, but still about 80% of the values of normal controls. However, the MCRs in high-dose clamp of two diabetic groups were not

Table 4

Metabolic clearance rate (MCR) of glucose for the last 30 min during the euglycemic clamp procedure at the low-dose (0–90 min) and the high-dose (90–180 min) insulin infusion

Group	MCRs for glucose (ml/kg/min)	
	Low-dose clamp	High-dose clamp
<i>Acute administration effect</i>		
Non-diabetic		
Saline treatment (7)	15.1 ± 0.6	50.3 ± 2.4
KJT treatment (6)	14.9 ± 0.7	51.4 ± 2.7
Diabetic		
Saline treatment (7)	6.7 ± 0.6 ^a	21.3 ± 1.7 ^a
KJT treatment (7)	12.3 ± 1.2 ^b	20.9 ± 1.3 ^a
<i>7-days administration effect</i>		
Non-diabetic		
Saline treatment (6)	15.8 ± 0.9	54.3 ± 3.0
Diabetic		
Saline treatment (6)	6.3 ± 0.5 ^a	21.6 ± 1.1 ^a
KJT treatment (6)	13.9 ± 1.0 ^b	26.0 ± 1.3 ^c
KJT treatment + L-NMMA (5)	6.9 ± 1.2 ^a	
Saline treatment + SNP (5)	14.0 ± 1.0	
KJT treatment + SNP (5)	13.7 ± 1.2	

Values are means ± S.E. The number of rats in each group is shown in parentheses.

^a Indicates a $P < 0.001$ vs. non-diabetic rats.

^b Indicates a $P < 0.001$ vs. saline treated diabetes.

^c Indicates a $P < 0.05$ vs. saline treated diabetes.

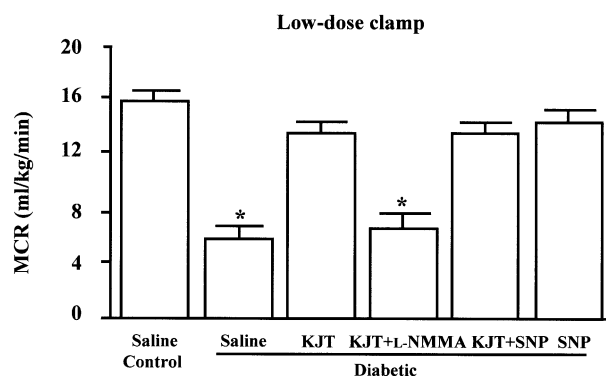


Fig. 1. 7-days administration effect: MCRs during the euglycemic clamp procedure in normal control and diabetes at the low-dose insulin infusion in the treatment with saline (normal and diabetes, $n = 6$, respectively), and KJT ($n = 6$), KJT + L-NMMA ($n = 5$), KJT + SNP ($n = 5$), and SNP ($n = 5$) treated diabetic rats, respectively. Data are expressed as the means $\pm$ S.E. * $P < 0.001$ vs. KJT treated diabetic and normal control.

significantly different. No effects of KJT treatment were seen on MCRs in normal rats during the low-dose or the high-dose insulin clamp (Table 4).

7-days administration effect of KJT on MCRs

During low-dose insulin clamp studies, the diabetes treated with 7-days KJT administration showed significantly higher MCRs than those of the diabetes given with saline ($P < 0.001$). Approximately 90% of the effect induced by 7-days administration with KJT in MCR was blocked by L-NMMA. No further additive effects on MCR were found in the KJT + SNP treated rats (Fig. 1). In high-dose insulin

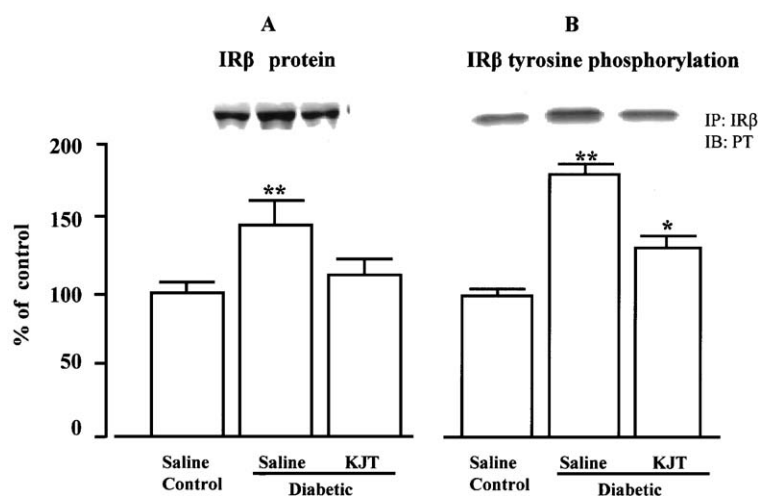


Fig. 2. Effect of 7-days administration of KJT on IRβ protein content (A) and tyrosine phosphorylation (B) in rat gastrocnemius muscle. Immunoprecipitation: IP; Immunoblotting: IB; Phosphotyrosine: PT. Data are expressed as the means $\pm$ S.E. for 5 rats, respectively. * $P < 0.05$ vs. controls; ** $P < 0.001$ vs. controls and KJT diabetic groups.

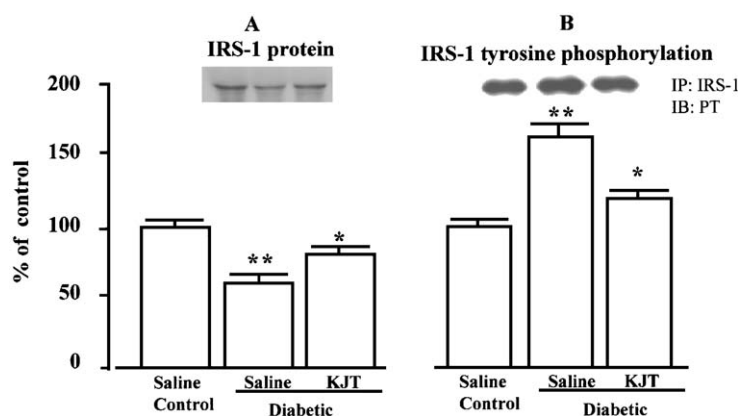


Fig. 3. Effect of 7-days administration of KJT on IRS-1 protein content (A) and tyrosine phosphorylation (B) in rat gastrocnemius muscle. Immunoprecipitation: IP; Immunoblotting: IB; Phosphotyrosine: PT. Data are expressed as the means $\pm$ S.E. for 5 rats, respectively. * $P < 0.05$ vs. saline diabetic groups; ** $P < 0.001$ vs. controls.

infusion, the MCR of KJT treated diabetes was higher than that of saline treated diabetes ($P < 0.05$), however, the MCRs of the above two diabetic groups were significantly less than the values of controls ($P < 0.001$; Table 4).

7-days administration effects of KJT on $IR\beta$, IRS-1, PI 3-kinase, and GLUT4 protein contents in rat skeletal muscle

The total amount of skeletal muscle $IR\beta$ protein (Fig. 2A) was significantly increased in diabetic rats compared with controls (140% of control, $P < 0.001$). Although, it did not recover to the normal control level after treatment and was indistinguishable between control and KJT treated diabetes ($P = 0.11$). In contrast, the IRS-1 protein (Fig. 3A) was decreased in diabetes compared with controls (59% of control, $P < 0.001$). However, it was increased to 75% of control level ($P < 0.05$) by KJT treatment. The total

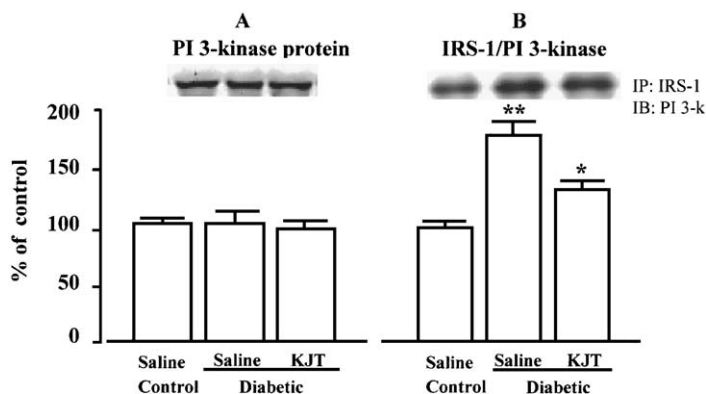


Fig. 4. Effect of 7-days administration of KJT on PI 3-kinase protein content (A) and the association of PI 3-kinase with IRS-1 (B) in rat gastrocnemius muscle. Immunoprecipitation: IP; Immunoblotting: IB. Data are expressed as the means $\pm$ S.E. for 5 rats, respectively. * $P < 0.05$ vs. saline diabetic groups; ** $P < 0.001$ vs. controls.

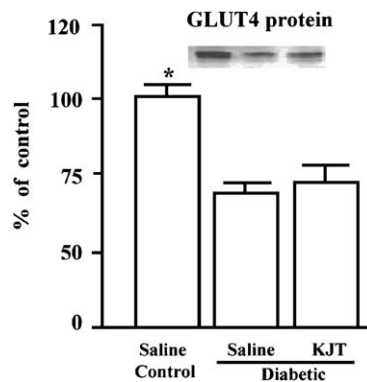


Fig. 5. Effect of 7-days administration of KJT on GLUT-4 protein content in rat gastrocnemius muscle. Data are expressed as the means $\pm$ S.E. for 5 rats. * $P < 0.001$ vs. saline and KJT treated diabetic groups.

amount of PI 3-kinase protein (Fig. 4A) was not significantly different among all experimental groups. GLUT-4 protein content in skeletal muscle of diabetic rats was significantly decreased compared with the controls (70% of control, $P < 0.001$), but, were not significantly different between KJT treated and saline treated diabetic rats (Fig. 5). No acute effect of KJT administration was observed in the protein content of IR β , IRS-1, PI 3-kinase, and GLUT4.

Effect of 7-days treatment with KJT on insulin-stimulated tyrosine-phosphorylation of IR β , IRS-1, and the IRS-1 association with PI 3-kinase in rat skeletal muscle

The tyrosine phosphorylation level of IR β was determined by immunoblotting the IR β antibody immunoprecipitates with the phosphotyrosine antibody. As shown in Fig. 2B, the tyrosine phosphorylation level of IR β in skeletal muscles of STZ-diabetes was significantly increased when compared to controls (186% of control, $P < 0.001$). However, a not so great increase was observed in KJT treated diabetes (135% of control, $P < 0.05$), indicating that KJT treatment was partly effective for correcting abnormal protein tyrosine phosphorylation in diabetes. The same trending was found in IRS-1 tyrosine phosphorylation. As shown in Fig. 3B, abnormal increase in IRS-1 tyrosine phosphorylation in STZ-diabetic rats (163% of control) was partly reversed by KJT treatment (120% of control). The IRS-1/PI 3-kinase association level was determined by immunoblotting the IRS-1 antibody immunoprecipitates with the PI 3-kinase antibody. As shown in Fig. 4B, IRS-1/PI 3-kinase association showed a significant increase in skeletal muscle of diabetic by 75% and 25% (saline and KJT treated diabetic, respectively) compared with controls ($P < 0.001$ and $P < 0.05$, respectively). The same improvement tendency in the tyrosine-phosphorylation of IR β , IRS-1, and the IRS-1/PI 3-kinase association was also observed in rats treated only one time with KJT (data not shown).

Discussion

The present study was undertaken to assess the effect of KJT on the insulin action in peripheral tissues. We investigated the insulin-induced glucose uptake in normal and STZ-diabetic rats using a two-

step hyperinsulinemic euglycemic clamp procedure. The MCR at low-dose insulin infusion primarily reflects the insulin sensitivity in peripheral tissues, these changes are thought to be caused mainly by changes in insulin receptor binding. High-dose insulin infusion leads to a maximal insulin action and responsiveness, predominantly indicating the capacity of post-receptor binding mechanisms [21,36]. The results of the euglycemic clamp study suggest that 1) impaired insulin sensitivity induced by STZ was improved by a single and 7-days administration of KJT, 2) 7-days treatment with KJT improved the decreased insulin responsiveness, however, not reaching the levels of controls, and 3) the KJT effect is probably mediated by NO.

The estimation of the whole-body insulin action with the euglycemic clamp technique reflects the glucose disposal in skeletal muscle (60–70%) [13], adipose tissue (about 10%), and liver (about 30%). Thus, an impaired insulin-stimulated glucose disposal during euglycemic clamp is mainly due to the reduced glucose uptake in skeletal muscle. However, insulin resistance induced by STZ-diabetes results from the impairment of insulin action, and do not only decreases peripheral glucose uptake, but also impairs the suppressing effect on hepatic glucose production (HGP) [7]. In the present study, high physiological insulin levels of about 180 pM achieved during low-dose insulin infusion rate reflects insulin sensitivity in peripheral tissues (i. e., skeletal muscle) and liver in the postprandial state. These levels of insulin are enough to inhibit the HGP in normal rats [45] but not in insulin-resistant high-fat-fed rats [34]. Therefore, we speculated that an inhibitory effect of KJT on HGP in the diabetic state can not be ruled out. The high-dose insulin infusion resulted in the pharmacological insulin range, which sufficiently suppresses HGP [45], therefore the impaired insulin responsiveness in peripheral tissues of diabetic rats might be improved by KJT treatment.

The skeletal muscle NOS activity is impaired during the progression of insulin-deficient diabetes [37]. Recently, it was reported that NO can contribute to insulin-mediated muscle glucose uptake independent of muscle blood flow [9], and glucose uptake in peripheral tissues *in vivo* is NO dependent [38]. Administration of L-NMMA *in vivo* results in the development of marked insulin resistance [40] and hyperglycemia [43], suggesting an important role for NO in the skeletal muscle glucose metabolism. L-NMMA administration to KJT-treated diabetes showed a decrease in the previously improved MCR (Fig. 1), suggesting that a consequent increase in NO production after KJT treatment might be an important factor to the effect of KJT on improving insulin resistance. On the other hand, exogenously administered NO, which is generated from the NO donor SNP, stimulates glucose transport in isolated skeletal muscle [6] by increasing GLUT4 levels at the cell surface [14]. Furthermore, SNP administration resulted in significantly increased glucose utilization in diabetic rats *in vivo* [25]. 7-days KJT treated diabetes were given SNP to determine whether KJT and SNP stimulate glucose uptake additively, resulting in no additional effect for combined KJT and SNP treatment compared with only SNP treated group (Fig. 1). Thus, we speculate that KJT and SNP improve insulin action in diabetic rats by the same metabolic pathway, by increasing NO production.

NO was reported to be involved in the regulation of skeletal muscle hemodynamics and glucose utilization through vasodilatation associated with insulin [46]. By means of thermography, Shikano and Niwa, 1987 [44] reported that KJT elicits a significant increase in the mean skin temperature in diabetic patients, appearing to have a strong vasodilative action. This raised us a question concerning whether the improvement of insulin action was caused only by an increase in blood flow or not. A previous study [26] showed that treatment with adenosine, which elicits vasodilatation, did not increase insulin action in STZ-diabetes, suggesting that enhanced blood flow alone can not improve insulin resistance induced by STZ. These evidences support, again, the hypothesis that KJT administration improves insulin action in

peripheral tissues via NO-related glucose uptake rather than increased blood flow or improved blood flow resulting from an increase in NO production. KJT administration is not able to produce even higher rates of glucose uptake in healthy animals via the NO pathway, i.e. the insulin sensitivity and responsiveness of normal rats could not be augmented over regular levels by a single dose of KJT administration. A previous study reported that KJT is effective in the diabetic state, but not in normal controls [22]. In the same fashion, C-peptide administration [26] or SNP infusion [25] could not improve the MCR of normal rats.

It is well known that the insulin signaling is very important for glucose metabolism. Recently, some studies suggested that IRS-1 plays a necessary role to couple signaling from the insulin receptor to PI 3-kinase and subsequent activation of the endothelial nitric oxide synthase (eNOS) [28]. PI 3-kinase plays a significant role in the insulin-signalling pathway related to the NO production [52]. Furthermore, studies provided direct evidence for a complete biochemical pathway involving the insulin receptor, IRS-1, PI 3-kinase, and eNOS that can account for important physiological actions of insulin to stimulate the production of NO in the vasculature [28,29,52,53]. Therefore, we further clarified the effects of KJT treatment on the molecular mechanisms of the *in vivo* insulin signaling in skeletal muscle obtained from the experimental rats. Studies on insulin signaling in skeletal muscle have demonstrated that STZ-diabetes causes an increase in the amounts of insulin receptors and total receptor phosphorylation, but a reduced efficiency of phosphorylation [17,41]. Our data show that 7-days KJT treatment significantly decreases the increased protein amount (Fig. 2A) and the increased tyrosine phosphorylation level of IR β (Fig. 2B).

IRS-1 plays a central role in insulin action for glucose transport in skeletal muscle [5,51]. Reduced IRS-1 expression was observed in Zucker fatty rats [4], and mice that are deficient in IRS-1 prepared by target gene-knockout exhibited hyperinsulinemia and glucose intolerance [5,47,51]. These findings suggest that decreased expression of the IRS-1 protein is involved in the development of insulin resistance. STZ-induced diabetes is associated with large increases in insulin-stimulated IRS-1 tyrosine phosphorylation [17,41] despite the decreased levels of total IRS-1 protein in skeletal muscle [41]. Our results show that KJT treatment significantly increases the amount of the IRS-1 protein (Fig. 3A) and inhibits the increased tyrosine phosphorylation level of IRS-1. However, these levels did not reach those of the controls (Fig. 3B).

The activation of PI 3-kinase is required for insulin and stimulate glucose transport [12,35]. In accordance with the findings of Laviola et al., 2001 [24], under our experimental conditions no difference in total PI 3-kinase protein amounts were observed in all studied groups (Fig. 4A). However, it was previously described that there is an increase in insulin-stimulated IRS-1-associated PI 3-kinase activity [15], in parallel with an increase in IRS-1/PI 3-kinase association in skeletal muscle of STZ-diabetes [15]. The present data indicate that KJT decreases the increased IRS-1/PI 3-kinase association (Fig. 4B), and this change appears to result in reversal of the abnormal activation of PI 3-kinase [15].

STZ-diabetes is characterized by reduction of muscle glucose uptake [27] and transport due to decreased GLUT4 content and insulin-induced transporter translocation [23,32] and impairment of the transporter intrinsic activity [32]. We found no effect of KJT on the regulation of GLUT4 protein content (Fig. 5). In our experiment, regardless we did not assess directly the effect of KJT on the GLUT4 translocation, we speculate that KJT improves insulin action via enhancing insulin signaling (i. e., ameliorated IR β , IRS-1 protein and efficiency of phosphorylation), consequently reversing NO production and the impaired GLUT4 translocation induced by STZ.

In summary, a single and 7-days administration with KJT improved insulin-stimulated skeletal muscle glucose uptake in STZ-diabetic rats. This improvement was associated with downregulation of insulin receptor protein content, and an increase in IRS-1 protein amounts, accompanied by changes in the response of the IR β and IRS-1 tyrosine phosphorylation, and the IRS-1 associated with PI 3-kinase. Increased NO production, induced by ameliorated tyrosine phosphorylation levels, might improved the impaired GLUT4 translocation without changing the GLUT4 protein content.

In conclusion, our study suggests that the improvement of impaired insulin action in STZ-diabetic rats by a single and repeated administration of KJT may be due, at least in part, to enhanced insulin signaling, and subsequent ameliorated production of NO. The present study supports the clinical usefulness of KJT for treating diabetic neuropathy.

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